Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

[CGPA-3.32] in Cycle 1

Project Guidelines:

Common Instructions

In the project documentation the sequence of preamble is

a) Title Page in specified format

b) Institute Certificate

c) Internship / Company Certificate

d) Guide certificate

e) Acknowledgement

f) Index with page numbers

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1. Application Development Project

Chapter No		Details
1		**Introduction**
	1.1	Company Profile
	1.2	Problem Statement
	1.3	Objectives of Proposed System
	1.4	Scope of the Proposed System
2		**Analysis and Design**
	3.1	Entity Relationship Diagram (ERD)
	3.2	Business Use Case Diagram
	3.3	Table Structure/Data Dictionary
3		**Testing**
	4.1	Test Cases with results
	4.2	Defect report / Test Log
4		**Limitations of Proposed System**
5		**Proposed Enhancements**
6		**References**
7		**User Manual (All input and output screens with consistent data)**

2. Cloud Computing

Project Requirement	**Details**
Cloud Environment Setup	Use AWS, Azure, GCP, or OpenStack
Service Model	Specify IaaS, PaaS, SaaS, or FaaS
Deployment Model	Public, Private, Hybrid, or Multi-Cloud

Index: Cloud Computing

3. Cyber Security:

Project Requirement	Details
Threat Analysis	Identify common and advanced threats (e.g., APTs, ransomware)
Security Controls	Implement network security, IAM, and data protection
Research Focus	Choose AI, IoT, Cloud, or Human Factors in Cybersecurity

Index: Cyber security

4. AI/ML/DL/Data Science

Project Areas

Students may choose projects in, but not limited to, the following areas: Machine Learning (ML) and Deep Learning (DL), Natural Language Processing (NLP), Computer Vision, Predictive Analytics, Big Data Analytics, Reinforcement Learning, AI-driven Automation, Data Visualization, AI Ethics and Responsible AI

Project Requirement	Details
Data Collection	Use real-world datasets or APIs
Model Development	Implement ML algorithms (e.g., regression, classification)
Evaluation	Use metrics like accuracy, precision, recall, and F1-score

Index: AI/ML/DL/Data Science

Chapter	Content	Page Number
Chapter 1	Introduction	
1.1	Problem Statement	
1.2	Objectives	
1.3	Dataset Description	
Chapter 2	Literature Review	
2.1	Existing Research	
2.2	Research Gaps	
Chapter 3	Methodology	
3.1	Data Preprocessing	
3.2	Model Selection	
3.3	Implementation	
Chapter 4	Results and Discussion	
4.1	Model Performance	
4.2	Visualization	
4.3	Insights	
Chapter 5	Conclusion	
5.1	Summary	
5.2	Future Work	
Chapter 6	References	
Chapter 7	Appendices	
Chapter 8	Annexure- Progress Sheet	

5. Guidelines for Entrepreneurship Project

Project Areas

Students aiming to foster an entrepreneurial ecosystem to become an entrepreneur with the essential skills, knowledge, and mindset required to identify market opportunities and successfully launch new ventures. Following are the guidelines to be followed by aspiring students for the Entrepreneurship development Projects:

Index: Entrepreneurship Development Project

6. AI Agenting

Project Areas

Students having project in may choose projects in, but not limited to, the following areas: Machine Learning (ML) and Deep Learning (DL), AI and creating its agents. Students undertaking projects in AI Agenting are required to follow the guidelines given below for effective planning, development, and implementation of their work.

Project Requirement	Details
Data Collection	Use real-world datasets or APIs
Model Development	Implement ML algorithms, AI agents (e.g., regression, classification)
Evaluation	Use metrics like accuracy, precision, recall, and F1-score, AI agents

Index: AI/ML/DL/Data Science/AI agents

Chapter	Content	Page Number
Chapter 1	Introduction	
1.1	Problem Statement	
1.2	Objectives	
1.3	Dataset Description	
1.4	Agentic System Overview (Describe agent-based architecture, Reasoning loop, autonomy and tool usage)	
Chapter 2	Literature Review	
2.1	Existing Research	
2.2	Research Gaps	
2.3	Agent based AI Systems (Architecture/Framework used for Autonoums agents and reasoning agents/systems)	
2.4	LLM Frameworks and Tools (Specific Technology used for agenting as LangChain, CrewAI, AutoGen, RAG etc)	
Chapter 3	Methodology	
3.1	Data Preprocessing	
3.2	Model Selection	
3.3	Agent architecture (Describe reasoning loop, planner and executer components)	
3.4	Prompt Engineering Strategy (Explain system prompts, role prompting and task instructions)	
Chapter 4	Implementation	
4.1	System Implementation	
4.2	Results (System Outputs)	
4.3	Tool Integration (Integration with APIs, databases, embeddings, retrieval pipelines)	
4.4	Knowledge Retrieval System (RAG) Explain vector databases, embeddings and database	

7. UIUX

Project Areas

The UI/UX Project Documentation Index outlines a structured approach for documenting the design and development of user-centred digital solutions. It guides students in presenting research, wireframes, prototypes, and usability considerations aligned with industry practices in User Experience Design and User Interface Design.

Project Requirement	Details
Data Collection	Analytics to understand user behavior and requirements in User Experience Design
Model Development	Develop user flows, wireframes, and high-fidelity prototypes applying principles of User Interface Design to create intuitive and user-friendly interfaces
Evaluation	Perform usability testing and evaluate using metrics like user satisfaction, task completion rate, and efficiency in User Experience Design

Index: UIUX

PROJECT REPORT

ON

NAME OF THE SYSTEM

FOR

NAME OF THE COMPANY

BY

NAME OF STUDENT

UNIVERSITY OF PUNE

MASTER IN COMPUTER MANAGEMENT

Maharashtra Education Society's
INSTITUTE OF MANGEMENT AND CAREER COURSES
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